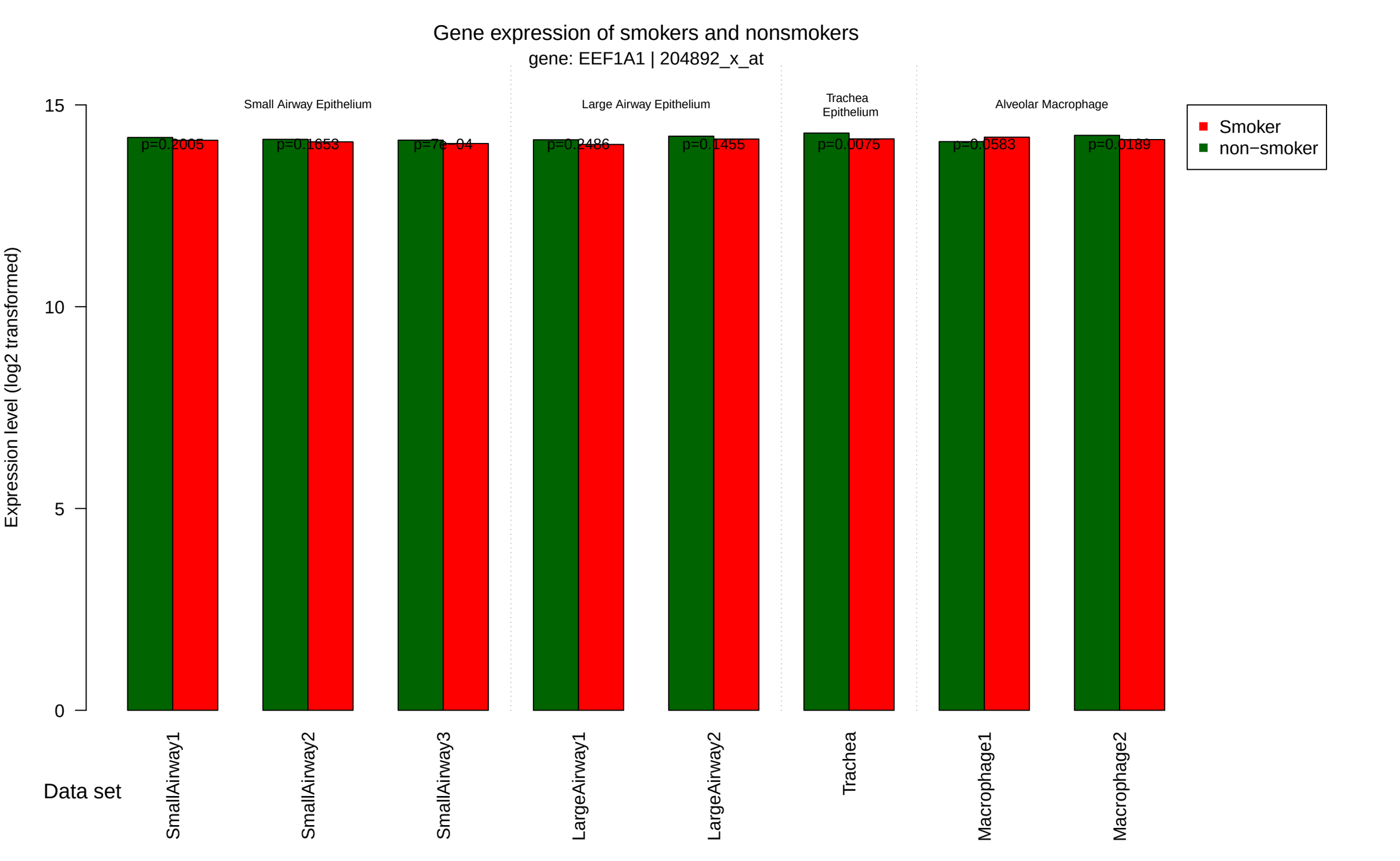
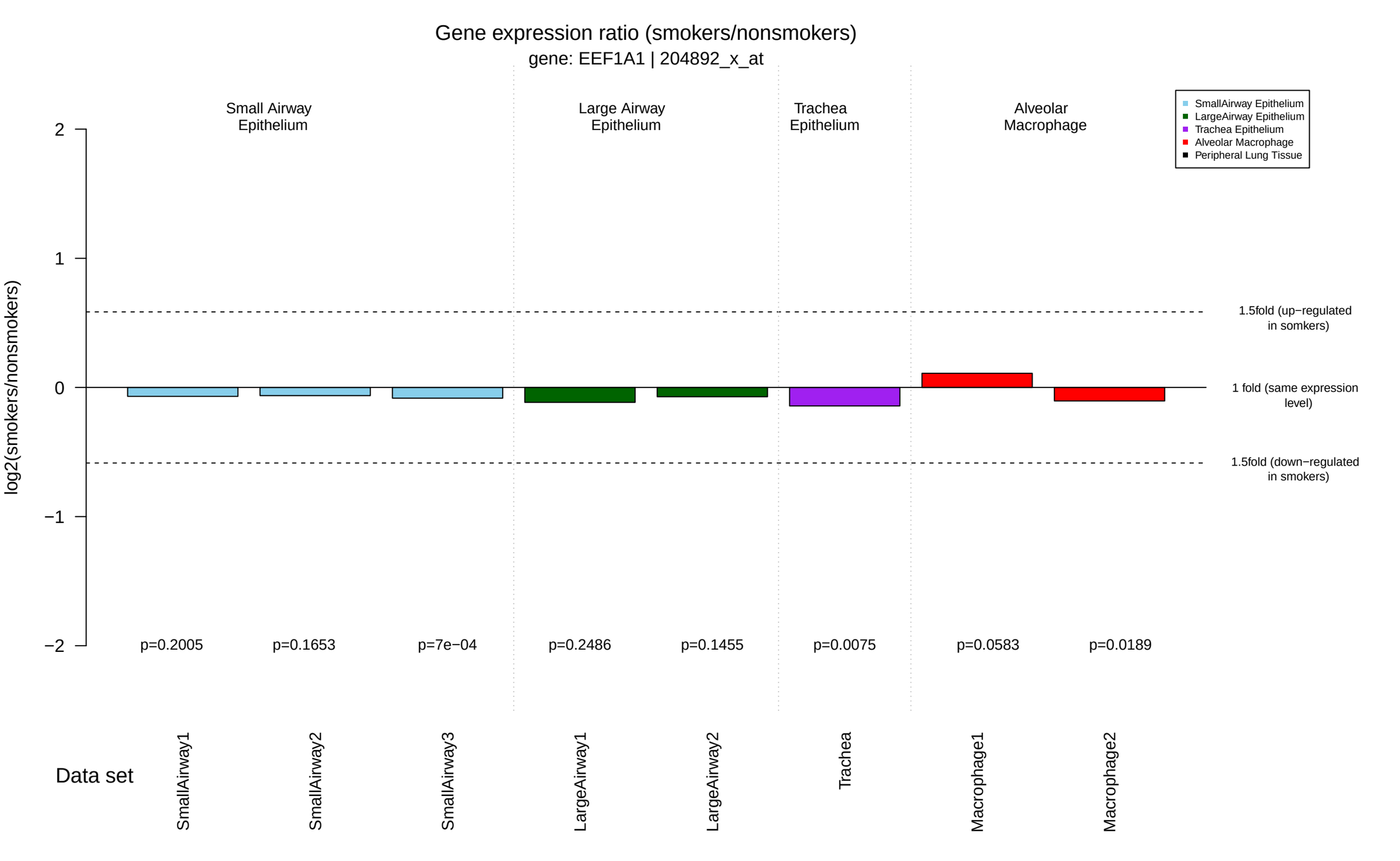
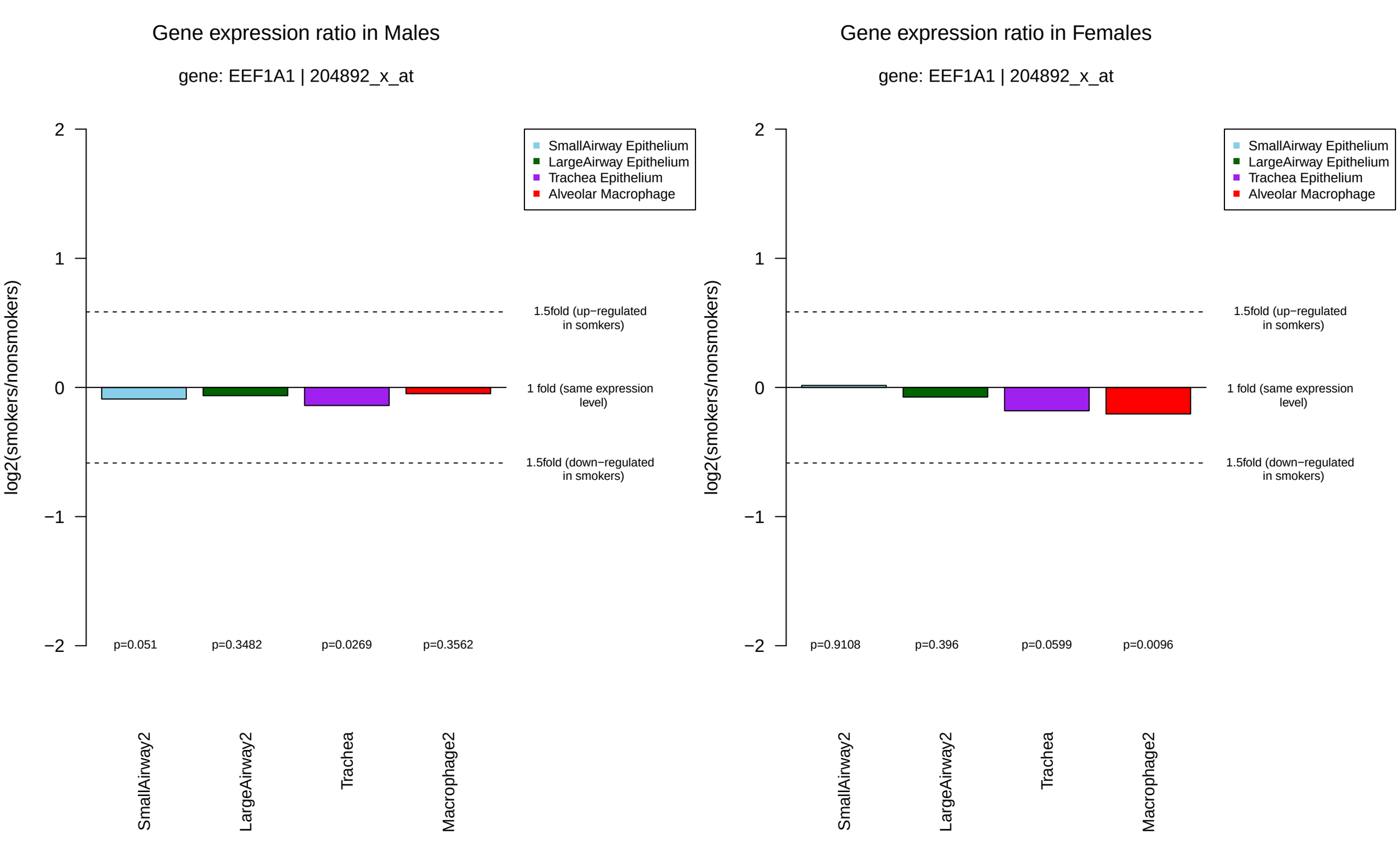
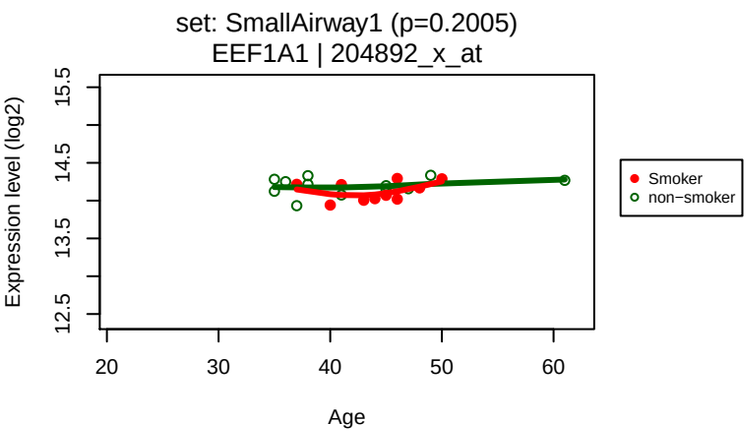
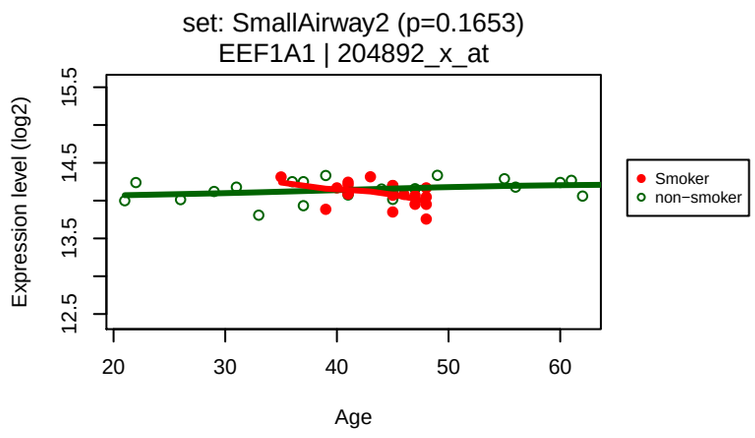
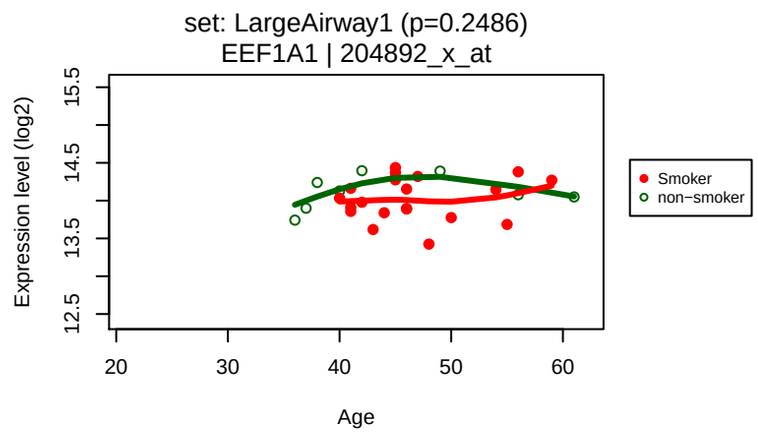

Gene expression of smokers and nonsmokers
gene: EEF1A1 | 204892_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2005
p=0.1653
p=7e−04
p=0.2486
p=0.1455
p=0.0075
p=0.0583
p=0.0189
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

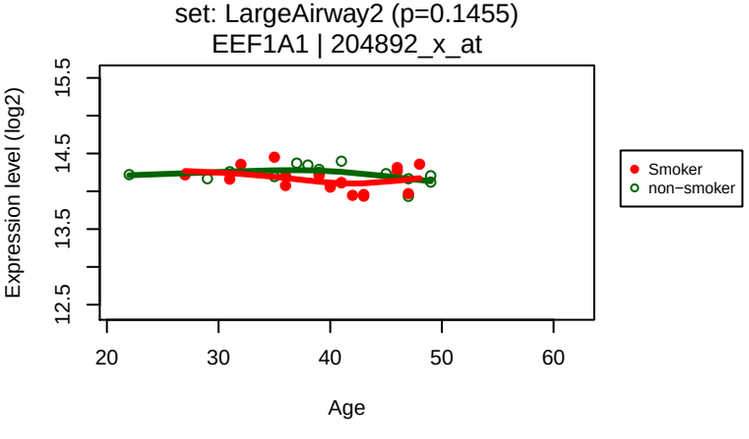

Gene expression ratio (smokers/nonsmokers)
gene: EEF1A1 | 204892_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2005
p=0.1653
p=7e−04
p=0.2486
p=0.1455
p=0.0075
p=0.0583
p=0.0189
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

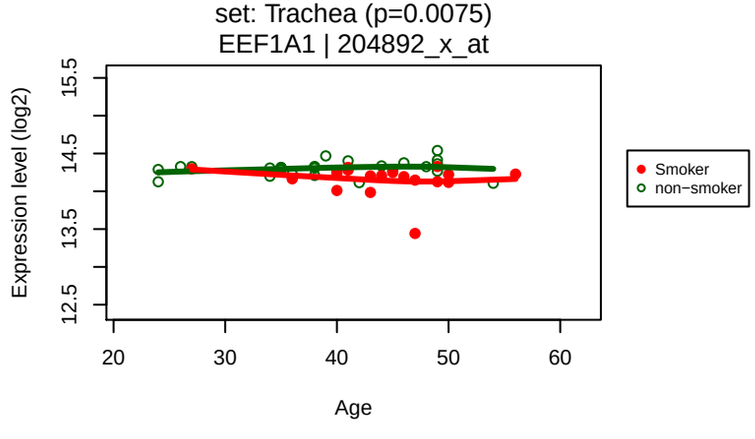

Gene expression ratio in Males
gene: EEF1A1 | 204892_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.051
p=0.3482
p=0.0269
p=0.3562
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: EEF1A1 | 204892_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9108
p=0.396
p=0.0599
p=0.0096
SmallAirway2
LargeAirway2
Trachea
Macrophage2

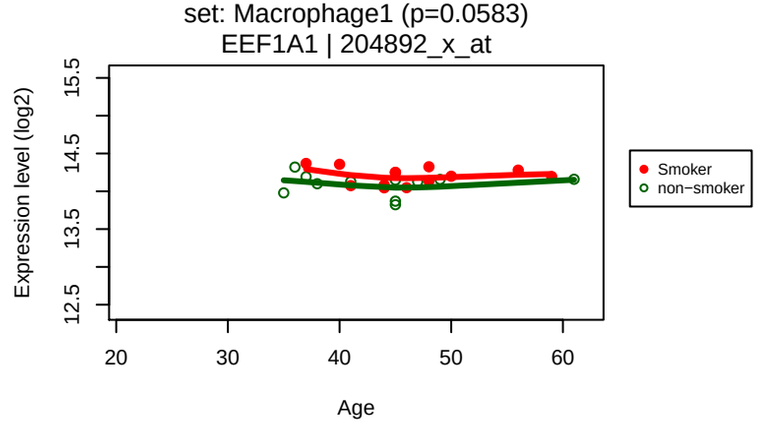

set: SmallAirway1 (p=0.2005)
EEF1A1 | 204892_x_at
Expression level (log2)
12.5
13.5
14.5
15.5
20
30
40
50
60
Age
Smoker
non−smoker

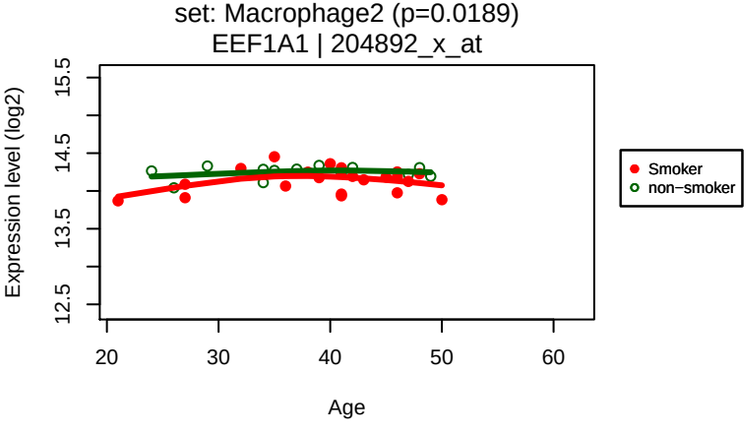

set: SmallAirway2 (p=0.1653)
EEF1A1 | 204892_x_at
Expression level (log2)
12.5
13.5
14.5
15.5
20
30
40
50
60
Age
Smoker
non−smoker

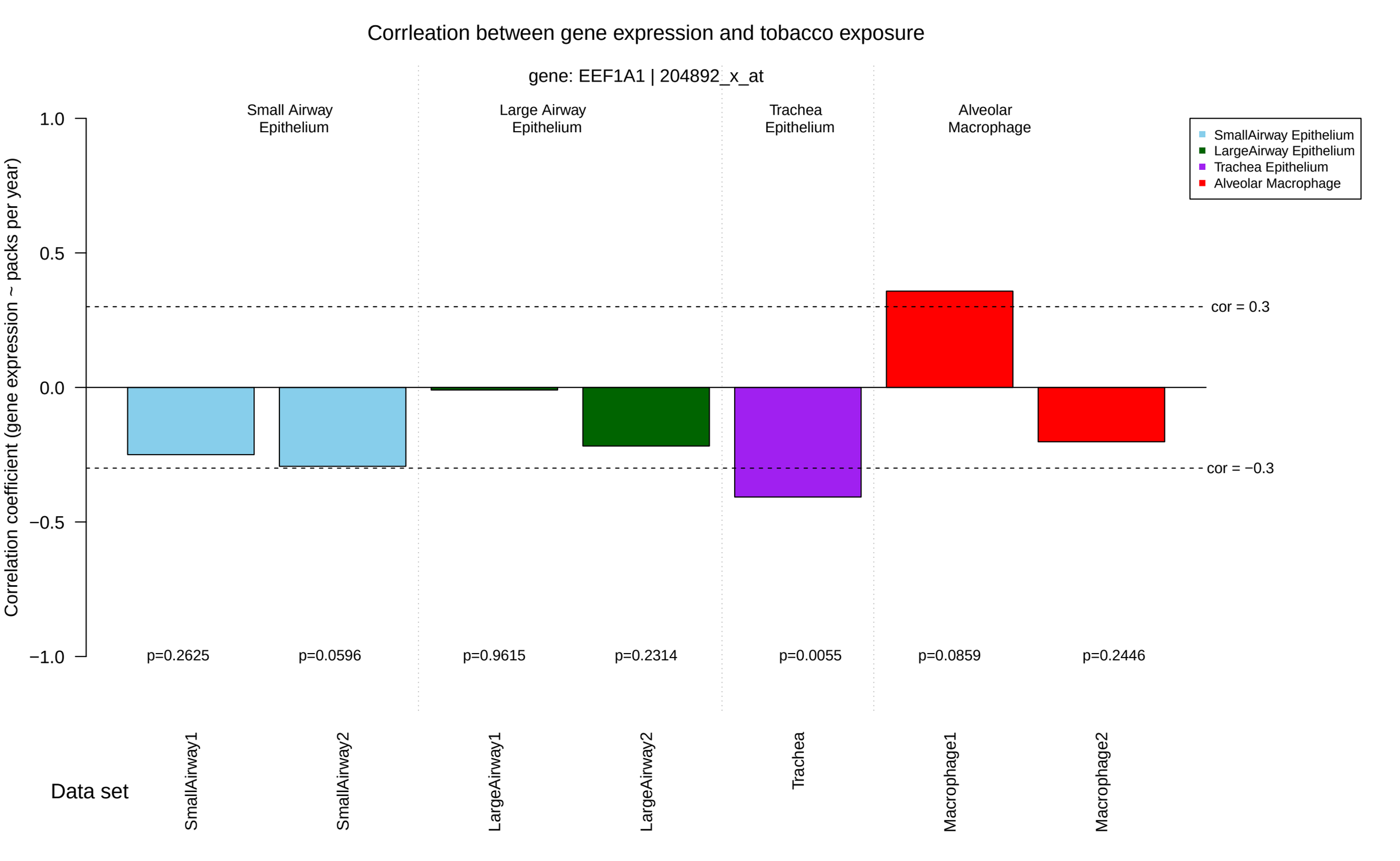

set: LargeAirway1 (p=0.2486)
EEF1A1 | 204892_x_at
Expression level (log2)
12.5
13.5
14.5
15.5
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1455)
EEF1A1 | 204892_x_at
Expression level (log2)
12.5
13.5
14.5
15.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.0075)
EEF1A1 | 204892_x_at
Expression level (log2)
12.5
13.5
14.5
15.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0583)
EEF1A1 | 204892_x_at
Expression level (log2)
12.5
13.5
14.5
15.5
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0189)
EEF1A1 | 204892_x_at
Expression level (log2)
12.5
13.5
14.5
15.5
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: EEF1A1 | 204892_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2625
p=0.0596
p=0.9615
p=0.2314
p=0.0055
p=0.0859
p=0.2446
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2